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LIBRARY MANAGEMENT SYSTEM

By

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1 INTRODUCTION

1.1 Background

Library Management System is an application which refers to library systems which are generally small or medium in size. It is used by librarian to manage the library using a computerized system where he/she can record various transactions like issue of books, return of books, addition of new books, addition of new students etc.

Books and student maintenance modules are also included in this system which would keep track of the students using the library and also a detailed description about the books a library contains. With this computerized system there will be no loss of book record or member record which generally happens when a non-computerized system is used.

All these modules are able to help librarians to manage the library with more convenience and in a more efficient way as compared Library Management System is a computerized system which helps user(librarian) to manage the library daily activity in electronic format. It reduces the risk of paperwork such as file loss, file damaged and time consuming. It can help users manage the transaction or record more effectively and timesaving to library systems which are not computerized.

1.2 Problem definition

The existing, paper-based manual registration system presents several issues that negatively impact on the library's efficiency. These problems include:

- File lost
- Space consumption
- Difficult to search for a record.
- Cost consuming
- A lack of real-time updates for member information.

1.3 Proposed solution

To effectively overcome the inefficiencies caused by the manual process, **library management system** is proposed. This new system will allow users to register for membership via the internet and will also provide library staff with an efficient way to manage member data.

1.4 Scope and context of the project (objectives)

- Improvement in control and performance

The system is developed to cope with the current issues and problems of library. The system can add users, validate users and is also bug free.

- Save cost.

After computerized system is implemented less human force will be required to maintain the library thus reducing the overall cost

- Save time.

Librarians can search for records by using few clicks of mouse and few search keywords thus saving his valuable time.

- Option of online Notice board

Librarian will be able to provide a detailed description of workshops going in the college as well as in nearby colleges.

- Lecture Notes

Teachers have the facility to upload lectures notes in a pdf file having size not more than 10mb.

1.5 Existing system

The current membership system at the library is fully **manual and paper based**. It strictly requires the physical presence of users at the library to register for new membership or to update their personal information.

1.6 Anticipated benefits

Implementing the Online Registration System is expected to deliver several key benefits:

- Faster registration processing times.
- Reduced paperwork and manual data handling.
- Improved accuracy in storing member data.
- Greater accessibility for users to register and update information.

1.7 Structure of the report

This final project report is structured into several main parts. It includes this introduction, followed by the methodology used, implementation details, presentation of results, and finally, the conclusion.

1.8 INTRODUCTION CONCLUSION

In conclusion, the implementation of the Library Management System with an Online Registration facility represents a significant step forward in modernizing the services of the Makandura Public Library. The proposed system successfully overcomes the challenges associated with the existing manual, paper-based registration process, such as long waiting times, difficulty in retrieving member information, and the risk of misplaced or inaccurate records. By introducing a fully digitized membership registration and management process, the system enhances overall operational efficiency and reduces the administrative burden on library staff.

Furthermore, the online platform provides a user-friendly and accessible solution for library members, allowing them to register, update their personal details, and access library services remotely without the need for repeated physical visits. This is particularly beneficial for students, researchers, working individuals, and senior citizens who may face time or mobility constraints. The system also ensures improved data accuracy, secure storage of information, and faster data retrieval, which collectively contributes to better decision-making and service delivery. Overall, the Library Management System not only improves the quality and efficiency of library operations but also supports the library's mission to serve the community more effectively by embracing modern technology and digital transformation.

2 METHODOLOGY

2.1 System design

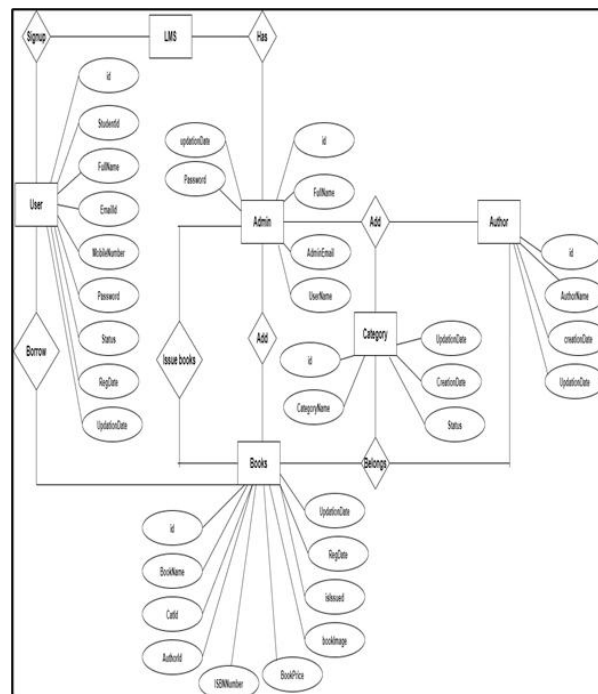
The Online Registration System was designed using a modular structure to ensure maintainability and clarity.

The system is fundamentally composed of three main modules: the **user interface**, the **database**, and the **admin panel**.

2.2 Database design

The database was designed to efficiently store and manage all necessary membership data. The structure includes separate tables to manage core system data, specifically: members, user accounts, and logs.

The database for the Library Management System was carefully designed to ensure efficient storage, easy retrieval, data integrity, and scalability. A relational database approach was used, where data is organized into separate but related tables to minimize redundancy and maintain consistency. The design follows normalization principles and clearly defined relationships between tables, allowing both library staff and system processes to manage information accurately and efficiently.



2.2- 1 : Entity Relationship Diagram

2.3 Database Tables

The system consists of the following main database tables:

2.3.1 Admin_Table

admin

Column	Type	Null	Default	Links to
id (<i>Primary</i>)	int(11)	No		
FullName	varchar(100)	Yes	NULL	
AdminEmail	varchar(120)	Yes	NULL	
UserName	varchar(100)	No		
Password	varchar(100)	No		
updateDate	timestamp	No	0000-00-00 00:00:00	

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	1	A	No	

2.3- 1:Admin Table

2.3.2 Table_authors

tblauthors

Column	Type	Null	Default	Links to
id (Primary)	int(11)	No		
AuthorName	varchar(159)	Yes	NULL	
creationDate	timestamp	Yes	CURRENT_TIMESTAMP	
UpdationDate	timestamp	Yes	NULL	

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	6	A	No	

2.3- 2: Tab Authors

2.3.3 Table_category

tblcategory					
Column	Type	Null	Default	Links to	
id (Primary)	int(11)	No			
CategoryName	varchar(150)	Yes	NULL		
Status	int(1)	Yes	NULL		
CreationDate	timestamp	Yes	CURRENT_TIMESTAMP		
UpdationDate	timestamp	No	0000-00-00 00:00:00		

Indexes									
Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment	
PRIMARY	BTREE	Yes	No	id	4	A	No		

2.3-3: Table_category

Table tblcategory track the record of category.

2.3.4 Books_Table

Table tblauthors track the details of authors.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	int(11)			No	None		AUTO_INCREMENT
2	BookName	varchar(255)	latin1_swedish_ci		Yes	NULL		
3	CatId	int(11)			Yes	NULL		
4	AuthorId	int(11)			Yes	NULL		
5	ISBNNumber	varchar(25)	latin1_swedish_ci		Yes	NULL		
6	BookPrice	decimal(10,2)			Yes	NULL		
7	bookImage	varchar(250)	latin1_swedish_ci		No	None		
8	isIssued	int(1)			Yes	NULL		
9	RegDate	timestamp			Yes	current_timestamp()		
10	UpdationDate	timestamp			Yes	NULL		ON UPDATE CURRENT_TIMESTAMP()

Indexes									
Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment	
PRIMARY	BTREE	Yes	No	id	8	A	No		

2.3-4: Books Table

2.3.5 Table_students

Table **tblstudents** for students record and student login details.

tblstudents

Column	Type	Null	Default	Links to
id (Primary)	int(11)	No		
StudentId	varchar(100)	Yes	NULL	
FullName	varchar(120)	Yes	NULL	
EmailId	varchar(120)	Yes	NULL	
MobileNumber	char(11)	Yes	NULL	
Password	varchar(120)	Yes	NULL	
Status	int(1)	Yes	NULL	
RegDate	timestamp	Yes	CURRENT_TIMESTAMP	
UpdationDate	timestamp	Yes	NULL	

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	7	A	No	
StudentId	BTREE	Yes	No	StudentId	7	A	Yes	

2.3- 5: Table_students

2.3.6 Issued_book_details

Books Table **tblissuedbookdetails** for maintain issued book and returned book record.

tblissuedbookdetails

Column	Type	Null	Default	Links to
id (Primary)	int(11)	No		
BookId	int(11)	Yes	NULL	tblbooks -> id
StudentID	varchar(150)	Yes	NULL	tblstudents -> StudentId
IssuesDate	timestamp	Yes	CURRENT_TIMESTAMP	
ReturnDate	timestamp	Yes	NULL	
RetrunStatus	int(1)	No		
fine	int(11)	Yes	NULL	

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	id	8	A	No	

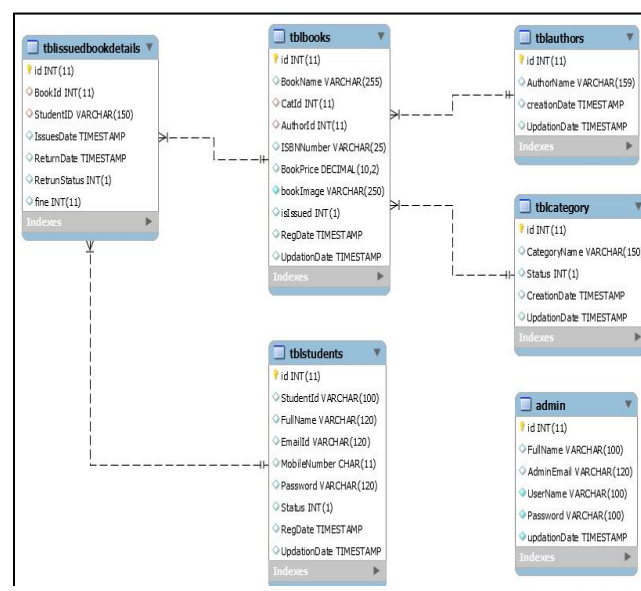
2.3- 6:Issued_book_details

2.4 Discussion of Database Design

The database design emphasizes data normalization, integrity, and efficiency. By separating data into multiple related tables, redundancy is minimized and consistency is maintained across the system. Primary keys uniquely identify each record, while foreign keys establish relationships between tables, ensuring referential integrity. This structured design allows the system to handle large volumes of data, support fast query execution, and generate accurate reports. Additionally, the modular nature of the database makes it flexible and scalable, allowing future enhancements such as fines management, reservations, or digital resource tracking to be easily integrated into the system.

2.5 ER Diagram Connections (Entity–Relationship Description)

Based on the implemented database schema, the Entity–Relationship (ER) diagram illustrates how the main entities of the Library Management System are logically connected. The relationships are designed to maintain data integrity, reduce redundancy,

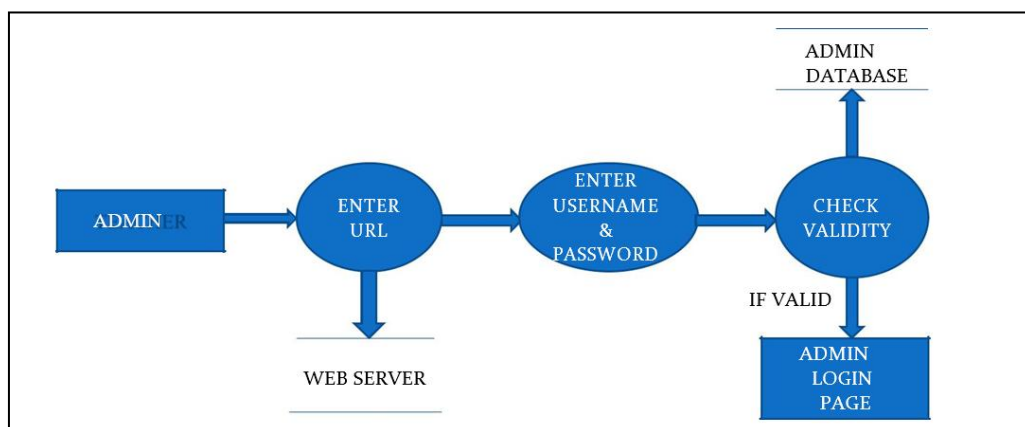


2.5- 1: Relationship between tables

and support efficient library operations.

2.6 Overall ER Design Explanation

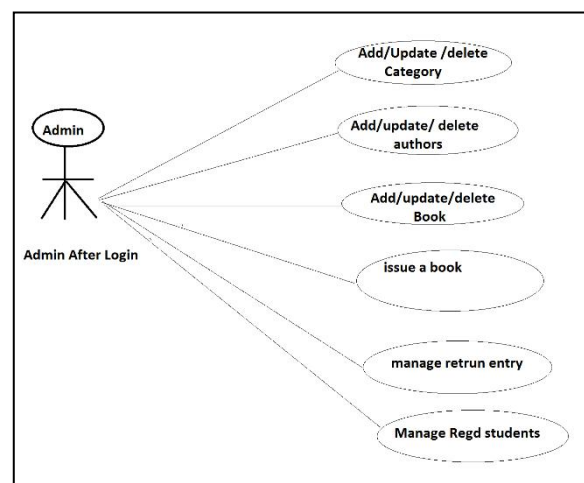
The ER diagram follows a normalized relational structure where each entity represents a real-world object in the library system. Primary keys uniquely identify records, while foreign keys establish meaningful relationships between entities. This design ensures referential integrity, supports efficient querying, and allows the system to scale easily for



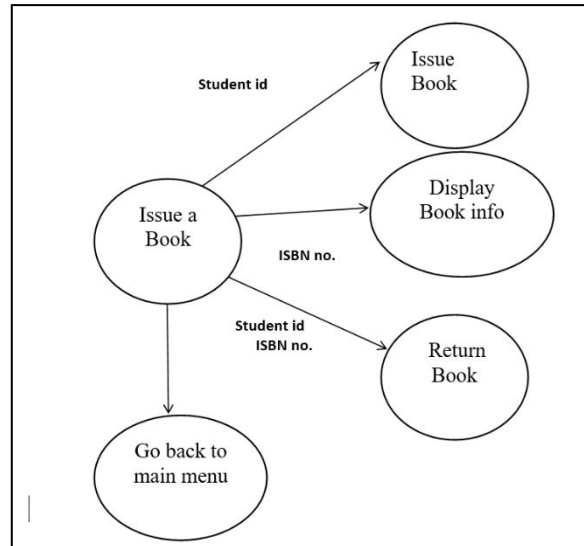
2.6- 1: Data Flow Diagram for Admin Login

future enhancements such as reservations, fines, or digital resource management.

After entering to the home page of the website , Admin can choose the ADMIN LOGIN option where they are asked to enter username & password , and if he/she is a valid user then a teacher login page will be displayed.



2.6- 2: Admin Data flow Diagram



2.6- 3: Data Flow Diagram for Issuing Books

2.7 Interface design (web)

The user interface and overall system interactions were designed for ease of use. The project utilized **HTML** and **CSS** for the frontend presentation, while **PHP** was employed for backend processing and scripting.

The screenshot shows the 'Online Library Management System' web interface. At the top, there is a navigation bar with links: HOME, USER LOGIN, USER SIGNUP, and ADMIN LOGIN. Below the navigation bar is a banner image of a library shelf. Underneath the banner is the 'USER LOGIN FORM'. The form contains a 'LOGIN FORM' header, an 'Enter Email id' input field, a 'Password' input field, a 'Forgot Password' link, and a 'Login' button. A link 'Not Register Yet' is also present next to the 'Login' button. The footer of the page displays '© 2022 Online Library Management System'.

2.7- 1: User Login Form



Online Library
Management System

HOME

USER LOGIN

USER SIGNUP

ADMIN LOGIN

USER SIGNUP

SIGNUP FORM

Enter Full Name

Mobile Number :

Enter Email

Enter Password

Confirm Password

Register Now

© 2022 Online Library Management System


Online Library Management System

[DASHBOARD](#)
[BORROWED BOOKS](#)
[ACCOUNT](#)

[LOG OUT](#)

USER DASHBOARD



8
Books Listed



0
Books Not Returned Yet



Issued Books

© 2022 Online Library Management System



Online Library
Management System

LOG OUT

DASHBOARD

ISSUED BOOKS

ACCOUNT

USER CHANGE PASSWORD

Change Password

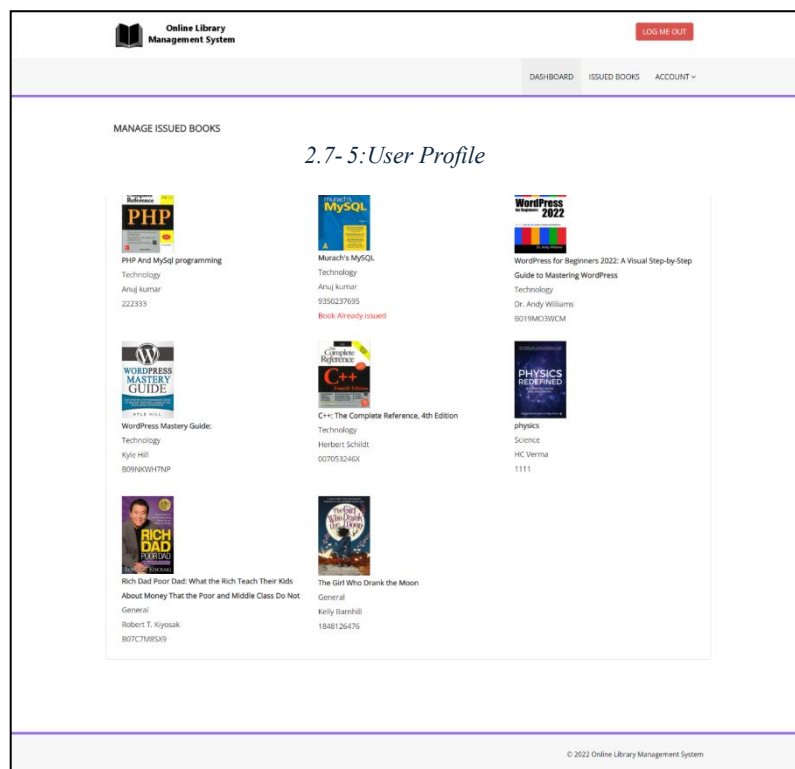
Current Password

Enter Password

Confirm Password

Change

© 2022 Online Library Management System



2.7- 6: Listed Books

2.8 Implementation Tools / language

The Library Management System is implemented using the following tools and technologies to ensure efficiency, reliability, and ease of maintenance.

2.8.1 Programming Languages

- **Front-End:**

- **HTML** – Used to design the structure of web pages ○ **CSS** – Used for styling and layout of the user interface ○ **JavaScript** – Used for form validation and interactive features.

- **Back-End:**

- **PHP** – Used to handle server-side logic such as user authentication, book management, and transaction processing.

2.8.2 Database

- **MySQL**

Used to store and manage data related to books, members, borrow/return records, and staff details. ○ Provides fast data retrieval and secure data handling.

2.8.3 Development Tools

- **XAMP**

Used as a local server environment including Apache and MySQL.

- **Web Browser (Google Chrome / Edge)**

Used for testing and accessing the system.

2.8.4 Operating System

- **Windows OS**

Used as the development and testing platform.

2.9 Testing

The aim of the system testing process was to determine all defects in our project .The program was subjected to a set of test inputs and various observations were made and based on these observations it will be decided whether the program behaves as expected or not. Our Project went through two levels of testing

1.Unit testing

2.integration testing

2.9.1 Unit Testing

Unit testing is undertaken when a module has been created and succesfully reviewed .In order to test a single module we need to provide a complete environment ie besides the module we would require

- The procedures belonging to other modules that the module under test calls
- Non local data structures that module accesses
- A procedure to call the functions of the module under test with appropriate parameters

Test For the admin module

- Testing admin login form-This form is used for log in of administrator of the system.In this we enter the username and password if both are correct administration page will open other wise if any of data is wrong it will get redirected back to the login page and again ask for username and password

- Student account addition- In this section the admin can verify student details from student academinc info and then only add student details to main library database it contains add and delete buttons if user click add button data will be added to student database and if he clicks delete button the student data will be deleted
- Book Addition- Admin can enter details of book and can add the details to the main book table also he can view the books requests .

Test for Student login module

- Test for Student login Form-This form is used for log in of Student .In this we enter thelibraryid, username and password if all these are correct student login page will open other wise if any of data is wrong it will get redirected back to the login page and again ask for libraryid, username and password.
- Test for account creation- This form is used for new account creation when student does not fill the form completely it asks again to fill the whole form when he fill the form fully it gets redirected to page which show waiting for conformation message as his data will be only added by administrator after verification.

2.9.2 Integration Testing

In this type of testing, we test various integration of the project module by providing the input. The primary objective is to test the module interfaces in order to ensure that no errors are occurring when one module invokes the other module.

2.10 METHODOLOGY CONCLUSION

In conclusion, the design and implementation of the Online Library Management System demonstrate a well-structured, efficient, and reliable solution to overcome the limitations

of the traditional manual library processes. By adopting a modular system design consisting of the user interface, database, and admin panel, the system ensures better maintainability, clarity, and scalability. The carefully planned relational database design, supported by normalized tables such as Author, Books, Publish, Users, and Borrow History, enables accurate data storage, fast retrieval, and strong referential integrity through clearly defined primary and foreign key relationships. The ER diagram further illustrates how real-world library entities are logically connected, allowing seamless tracking of books, users, authors, and borrowing activities while minimizing redundancy and data inconsistencies.

Additionally, the web-based interface designed using HTML, CSS, JavaScript, and PHP provide a user-friendly and interactive environment for both library users and administrators, making the system easy to use and accessible. The use of reliable implementation tools such as MySQL, WAMP, and modern web browsers ensures stable performance and efficient data handling. Comprehensive testing procedures, including unit testing, integration testing, system testing, user acceptance testing, and basic security testing, confirm that the system functions accurately, securely, and according to user requirements. Overall, this Library Management System successfully integrates sound system design principles, robust database architecture, effective implementation technologies, and thorough testing practices, resulting in a dependable and modern solution that enhances library operations and improves service delivery to the community while also providing a strong foundation for future system enhancements.

3 RESULTS AND DISCUSSION

3.1 Achievements

The project successfully achieved its core objectives by delivering a functional Online Registration System. A significant achievement was the system's capability to successfully handle **online registration** and subsequent **profile updates** for library members. This directly addresses the goal of digitizing the membership process and providing an efficient data management tool for the library staff.

3.2 Feedback from users before and after the product deployed

While the report does not explicitly detail feedback *prior* to deployment, post deployment evaluation showed positive user response. Users reported improved system performance, specifically noting the **improved speed and convenience** offered by the new online platform, confirming the anticipated benefit of faster registration and improved accessibility.

3.3 Problem refinements

During the testing and evaluation process, minor issues were identified. Specifically, minor **UI inconsistencies** were corrected and refined. This step ensured that the user interface was highly polished and contributed to a smoother user experience.

3.4 Further refinements

Based on the results and evaluation, a key area identified for future enhancement pertains to security and data verification. A specific refinement proposed for immediate future work is the implementation of **email verification** during the registration process.

3.5 Additional areas to be considered

Beyond the proposed email verification, the successful implementation of the system opens the door for broader system expansions. Additional areas to be considered for comprehensive future work include the integration of features such as **SMS alerts** and developing dedicated **mobile app support** to further enhance accessibility and member services.

3.6 RESULTS AND DISCUSSION CONCLUSIONS

In conclusion, the successful completion of the Online Registration System marks a significant milestone in achieving the overall objectives of the Library Management System project. The system effectively digitized the membership registration process, enabling library members to register online and update their profiles with ease, while providing library staff with an efficient and reliable data management tool. Positive feedback received after deployment highlights the system's improved performance, speed, and convenience, demonstrating its ability to enhance user satisfaction and accessibility compared to the previous manual process. Throughout the testing and evaluation stages, minor issues such as user interface inconsistencies were identified and carefully refined, resulting in a more polished and user-friendly interface that supports smooth system interaction. Furthermore, the evaluation process helped identify important areas for future refinement, particularly in strengthening system security through features

such as email verification during user registration. The project also lays a strong foundation for future enhancements, including the integration of SMS notifications and the development of mobile application support, which would further expand accessibility and improve communication with users. Overall, this phase of the project confirms that the system not only meets its initial goals but also provides a scalable and adaptable platform capable of evolving to meet future technological and user needs.

4 CONCLUSION AND FURTHER WORK

4.1 Conclusion

This website provides a computerized version of library management system which will benefit the students as well as the staff of the library. It makes entire process online where student can search books, staff can generate reports and do book transactions. It also has a facility for student login where student can login and can see status of books issued as well request for book or give some suggestions. It has a facility of teacher's login where teachers can add lectures notes and also give necessary suggestion to library and also add info about workshops or events happening in our college or nearby college in the online notice board. There is a future scope of this facility that many more features such as online lectures video tutorials can be added by teachers as well as online assignments submission facility , a feature Of group chat where students can discuss various issues of engineering can be added to this project thus making it more interactive more user friendly and project which fulfills each users need in the best way possible. Although the system successfully fulfills its current requirements, there is considerable potential for future enhancements to further improve its functionality, security, and user experience. One key area for further development is the implementation of stronger security measures, such as email and SMS verification, role-based access control, and encrypted data storage to ensure the protection of sensitive user information. Additionally, integrating automated notifications for membership renewals, book availability, and due date reminders would enhance user engagement and service efficiency. The development of a mobile application could further extend accessibility, particularly for users who primarily access services via smartphones. Advanced reporting and analytical features could also be incorporated to assist library administrators in monitoring usage patterns, managing resources, and supporting informed decision making. With these future improvements, the Library Management System can evolve into a fully comprehensive digital platform that effectively supports the long-term operational and strategic needs of the library.

5 References

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